AirQI-LLM:

Air Quality Analysis

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Abstract

The AirQI-LLM project introduces a sophisticated solution to the growing challenge of environmental data accessibility. While high-resolution air quality data is readily available through organizations like the United States Environmental Protection Agency (EPA), the technical complexity of pollutant metrics often prevents the general public from making informed health decisions. This project bridges that gap by integrating Supervised Machine Learning with Large Language Models (LLMs) through a Retrieval-Augmented Generation (RAG) architecture.

The system serves a dual purpose: first, it employs regression algorithms, specifically Random Forest and Gradient Boosting, to provide accurate Air Quality Index (AQI) predictions based on historical and real-time environmental parameters. Second, it utilizes a localized knowledge base stored in a ChromaDB vector store to power an interactive chatbot. This allows users to receive context-aware, natural language explanations regarding health risks and pollutant causes, ensuring the AI's responses are grounded in scientific fact rather than generative hallucination.

Implemented using a modern stack including Python, LangChain, and Streamlit, the AirQI-LLM prototype transforms raw numerical data into actionable intelligence. Initial results demonstrate that the combination of predictive modeling and conversational AI significantly improves the interpretability of environmental trends. Future iterations aim to incorporate real-time API streaming and cloud-based deployment to offer a scalable tool for public health safety across the United States.

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1. Introduction

This document outlines the project plan for AirQI-LLM, an innovative Air Quality Analysis Tool designed for the United States of America. The system leverages Large Language Models (LLMs) and Machine Learning (ML) to provide comprehensive, interpretable, and user-friendly insights into air quality conditions.

AirQI-LLM aims to bridge the gap between complex environmental datasets and public understanding by offering natural language explanations, context-aware responses, and an interactive AI-driven experience related to air quality and its health implications.

2. Problem Statement: Bridging the Air Quality Information Gap

The project addresses the following major challenges:

- Public Awareness Challenge: Citizens often lack easy access to clear and understandable air quality insights.
- Data Complexity: Air quality data contains multiple pollutants and meteorological parameters that are difficult for non-experts to interpret.
- Limited Interpretability: Traditional ML models generate predictions but fail to explain *why* air quality changes.

3. Project Objectives

The key objectives of the AirQI-LLM project are:

- Develop an Air Quality Index (AQI) Prediction System
- Build an Interactive LLM-based Chatbot
- Utilize USA Air Quality datasets
- Implement Retrieval Augmented Generation (RAG) for accurate and grounded responses
- Provide real-time or historical AQI insights
- Integrate Machine Learning regression models
- Store environmental knowledge as structured documents
- Design a simple and intuitive user interface
- Allow users to ask natural language questions

4. Data Sources

The primary dataset is obtained from the United States Environmental Protection Agency (EPA). The data is provided in CSV format and includes historical air quality measurements.

Key Pollutants Used :

- PM2.5
- PM10
- NO_2
- SO_2
- CO
- O_3

us_air_quality_sample.csv

Optional meteorological features such as temperature and humidity may also be integrated to improve prediction accuracy.

5. Building the Knowledge Base

The knowledge base forms the foundation of the RAG system and includes:

- AQI standards and classification ranges
- Health impact descriptions
- Pollutant explanations
- Model and system documentation

The knowledge is stored as Markdown (.md) and Text (.txt) files to ensure flexibility and ease of ingestion.

6. System Architecture: RAG Pipeline Flow

The system architecture integrates RAG with Machine Learning models.

Component	Description
Document Loader	Loads all knowledge files and converts them into document
Text Splitter	Splits large documents into manageable chunks while preserving context
Embedding Model	Converts text chunks into numerical vectors using HuggingFace models
Vector Store	ChromaDB stores embeddings and enables similarity search
Retriever	Retrieves relevant document chunks based on user queries
LLM	Generates accurate responses grounded in retrieved context
User Interface	Enables interactive user engagement

6.1 RAG Process Flow

1. User Query: e.g., “Is air quality in Chicago safe today?”
2. Retrieval: Relevant document chunks are fetched from ChromaDB
3. Context Injection: Retrieved chunks are passed to the LLM
4. Grounded Response: The LLM generates an accurate and explainable answer

6.2 System Prompt Design

The system prompt is:

- Domain-focused on air quality
- Strictly grounded in retrieved context
- Configured to politely refuse responses if data is unavailable

```
User Interface (Gradio / Streamlit)
          |
          v
     User Query
          |
          v
       Retriever
          |

     ChromaDB
   (Vector Store)
          |
          v
   Retrieved Context
          |
          v
     LLM (GPT / LLaMA)
          |
          v
   Natural Language Response
```

ML Pipeline(Parallel Flow):

Dataset -> Preprocessing -> ML Model -> AQI Prediction

7. Machine Learning Model

7.1 Learning Type and Algorithms Used

The ML component uses Supervised Learning (Regression) techniques.
Algorithms implemented:

- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor

Target Variable:

- Air Quality Index (AQI)

7.2 ML Pipeline

1. Load dataset
2. Data cleaning and preprocessing
3. Feature selection
4. Train-test split
5. Model training
6. Model evaluation
7. Model persistence (saving the trained model)

7.3 Model Evaluation Metrics

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 Score

8. Current Prototype and Limitations

8.1 Current Prototype Status

- Jupyter Notebook-based implementation
- Local ChromaDB vector store
- Trained ML regression model

8.2 Prototype Limitations

- No persistent chat history
- Manual data ingestion
- Local deployment only

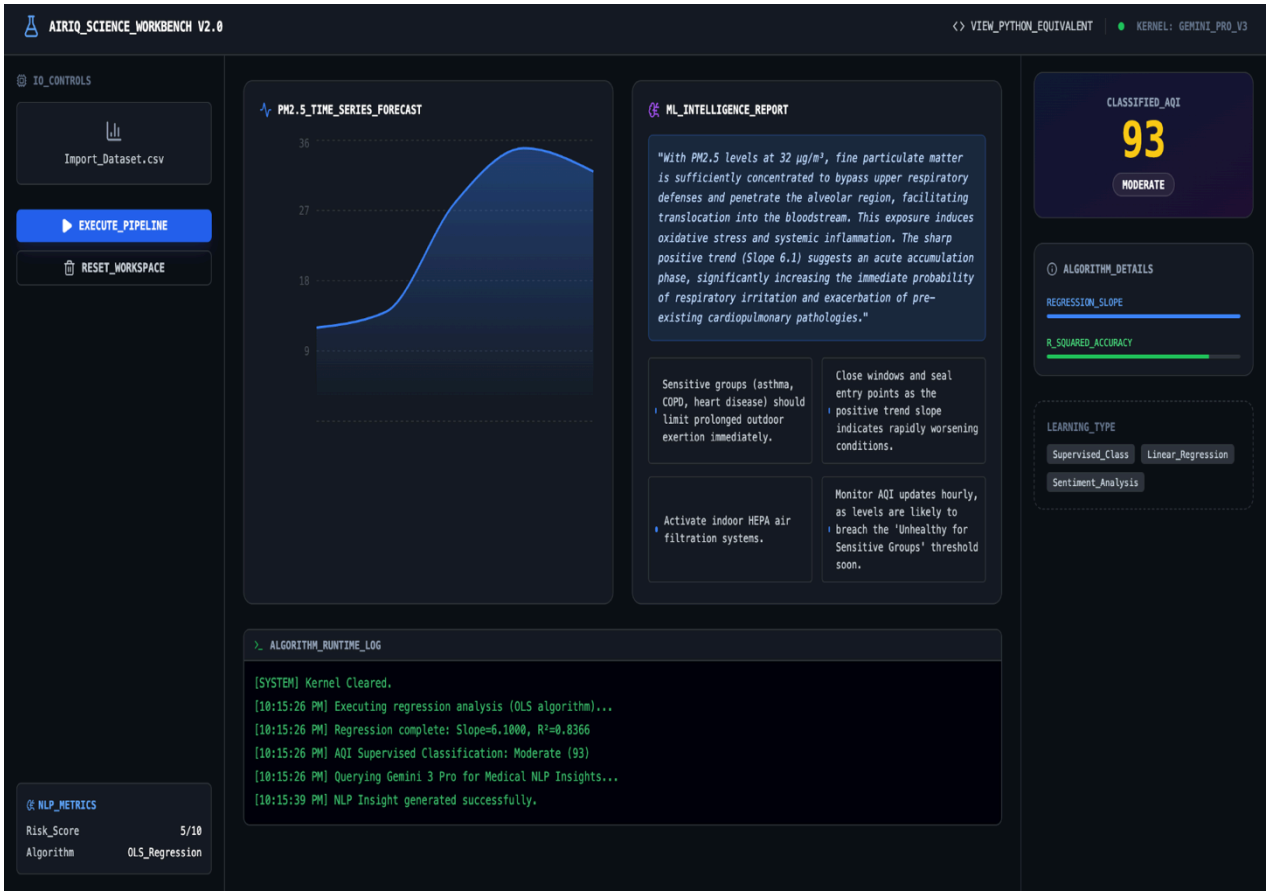
9. Future Developments

Planned enhancements include:

- Modular and scalable codebase
- Chat memory integration
- Cloud deployment
- Real-time AQI API integration
- Web and mobile application development

10. AirQI-LLM Tool Features

- Upload CSV datasets
- Predict AQI values
- AI-powered chat for air quality queries
- Pollutant trend visualization
- Health-based recommendations



10.1 User Interface Flow

1. User uploads dataset
2. User clicks Predict AQI
3. Results are displayed
4. User interacts with chatbot

10.2 Tool Architecture

Component	Technology
Frontend	Gradio / Streamlit
Backend	Python
ML	Scikit-learn
LLM	LLaMA / GPT via LangChain

10.3 Sample Questions the Tool Can Answer

- What does PM2.5 mean?
- Is an AQI of 160 dangerous?
- Why is ozone high today?
- Predict AQI for New York.

11. Technologies Used

- Python
- Pandas, NumPy
- Scikit-learn
- LangChain
- HuggingFace
- ChromaDB
- Gradio / Streamlit

12. Conclusion

The AirQI LLM project presents an intelligent, interpretable, and user-centric solution for air quality analysis in the USA. By combining Machine Learning predictions with LLM powered explanations through a Retrieval Augmented Generation framework, the system not only predicts AQI but also enhances public understanding of environmental health risks.

This project demonstrates how AI can bridge the gap between complex scientific data and everyday users, contributing toward improved awareness, informed decision making, and public health safety. With future scalability and real-time integration, AirQI LLM has strong potential for real-world deployment.

13. References

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